Tejas Shivakumar Kasetty

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Profile

Graduate Researcher at Mila – Quebec AI Institute, with research interests in continual and open-ended learning, compositional representations, adaptive curricula and data selection, and information-theoretic perspectives on learning and generalization.

Education

2022–2026 **M.Sc. in Computer Science** Montreal, Canada
Université de Montréal / Mila - Québec AI Institute
Co-supervised by Guillaume Lajoie and Dhanya Sridhar. GPA: 4.2/4.3.

2014–2018 **B.Tech. in Computer Science** Bengaluru, India
PES University
Undergraduate thesis and research internship in computer vision at the Center for Pattern Recognition and Machine Intelligence (CPRMI).

Publications

2026 **Defining and Generating Compositional Structure**
Eric Elmoznino, **Tejas Kasetty**, Blaise Agüera y Arcas, Yoshua Bengio, Guillaume Lajoie
arXiv preprint; formalizes compositional structure and uses it to generate compositional data, with implications for curriculum design

2026 **Simplifying the Modeling of Arbitrary Conditionals in Natural Language**
Yinhan Lu*, Eric Elmoznino*, Léo Gagnon, Sarthak Mittal, **Tejas Kasetty**, Guillaume Lajoie
arXiv preprint; introduces AC-GPT for arbitrary conditional language modeling

2026 **A Compression Perspective on Simplicity Bias**
Tom Marty, Eric Elmoznino, Leo Gagnon, **Tejas Kasetty**, Mizu Nishikawa-Toomey, Sarthak Mittal, Guillaume Lajoie, Dhanya Sridhar
arXiv preprint

2025 **In-context Learning and Occam's Razor**
Eric Elmoznino*, Tom Marty*, **Tejas Kasetty**, Leo Gagnon, Sarthak Mittal, Mahan Fathi, Dhanya Sridhar, Guillaume Lajoie
International Conference on Machine Learning (ICML), poster

2025 **Next-Token Prediction Should be Ambiguity-Sensitive: A Meta-Learning Perspective**
Leo Gagnon, Eric Elmoznino, Sarthak Mittal, Tom Marty, **Tejas Kasetty**, Dhanya Sridhar, Guillaume Lajoie
ICML Workshop on Efficient Systems for Foundation Models; arXiv preprint

2024 **Evaluating Interventional Reasoning Capabilities of Large Language Models**
Tejas Kasetty*, Divyat Mahajan, Gintare Karolina Dziugaite, Alexandre Drouin, Dhanya Sridhar
NeurIPS CALM Workshop, poster

Research Experience

Sept 2024–Present **Graduate Student Researcher** Montreal, Canada
Mila - Quebec AI Institute

- Researching compositionality and compositional generalization in neural learning systems as part of Guillaume Lajoie's lab.
- Developing data-ordering and curriculum-learning methods for better representations, using autoregressive/generative models to evaluate predictive efficiency.
- Building PyTorch pipelines for policy learning over image/patch orders and downstream representation evaluation.

July 2023–Aug 2024 **Research Intern** Montreal, Canada
Mila - Quebec AI Institute

- Evaluated causal and interventional reasoning abilities of large language models under controlled benchmark settings, separating memorization from reasoning.
- Contributed to theory and experiments connecting in-context learning as meta-learning to Occam-style simplicity bias.

May 2017–April 2018 **Research Intern** Bengaluru, India

Center for Pattern Recognition and Machine Intelligence (CPRMI) Lab, PES University

- Worked on computer vision research for face recognition and image deblurring during a summer internship and undergraduate thesis.
- Implemented neural-network and image-processing methods for degraded visual data and restoration-quality evaluation.

Industry Experience

May 2023–Nov 2023 **Machine Learning Developer Intern** Montreal, Canada

Intellisports Inc.

- Built sequence-modeling approaches for human activity recognition from inertial measurement unit (IMU) time-series data.
- Developed models and evaluation pipelines for activity recognition and repetition counting in sports and fitness settings.

June 2018–Jan 2020 **Software Development Engineer - Big Data and ML** Bengaluru, India

MiQ

- Implemented Spark-based libraries and services supporting scalable ML workflows on large-scale advertising and analytics data.
- Helped analysts and data scientists automate data pipelines and train machine-learning models in distributed computing environments.

Jan 2018–June 2018 **Software Development Intern - Full Stack** Bengaluru, India

MiQ

- Developed a web application for creating, running, and scheduling workflows.
- Refactored backend services from a monolithic architecture to microservices and migrated deployment to Kubernetes.

Selected Projects

2023–Present **Quick-Recipe: Knowledge Extraction from Cooking Videos** Montreal, Canada
Université de Montréal

- Explored multimodal text-video models for extracting objects, actions, and procedural knowledge from cooking videos.
- Used temporal segment models and attention mechanisms for text-to-video temporal grounding.

2022 **NHL Analytics** Montreal, Canada
Université de Montréal

- Built an end-to-end pipeline for NHL play-by-play data from 2018–2021, including querying, visualization, and live-game analytics.
- Trained an ensemble model for shot/miss/goal prediction and deployed a web application for real-time game insights.

Technical Skills

Languages Python, Scala, Java, C, C++, MATLAB, LaTeX
ML / Research PyTorch, PyTorch Lightning, Hydra, TensorFlow, scikit-learn, NumPy, Pandas, Transformers, autoregressive models, reinforcement learning, causal inference, representation learning
Systems / Tools Spark, Git, Jupyter, Databricks, Qubole Notebooks, Kubernetes

Awards, Service, and Leadership

2019 **Best Newcomer Award**, MiQ, Bengaluru, India.
2023–2024 **Student Affairs Committee Member, Content Manager**, UNIQUE, Montreal. Supported student programming at the intersection of neuroscience and AI in Quebec.
2023 **Organizer / Volunteer**, UNIQUE Student Symposium. Assisted with event logistics for a student research symposium in neuroscience and AI.
2014–2017 **PES University Football Team**. Represented the university in multiple tournaments.